

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	39.0 / Female
Department	General Medicine
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Urine

Clinical Presentation

Chief Complaints: Burning urination;Pelvic pressure

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9050.0	cells/μL	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	9.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.1	mg/dL	3.5 - 7.2
Blood Urea	25.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Streptococcus agalactiae
Bacteria Type	Gram-positive coccus (Group B Streptococcus; significant in pregnancy)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 480 mg IV over 30 minutes daily for a 80 kg patient)

Precautions: Check baseline creatine phosphokinase (CPK) and repeat weekly; discontinue if CPK >5× ULN. Renal function is normal (creatinine 0.7 mg/dL) so standard dosing is appropriate. Monitor for hypersensitivity reactions. Not indicated for pulmonary infections.

Rationale: Streptococcus agalactiae (Group B Streptococcus) is a Gram-positive coccus that is highly susceptible to daptomycin, the only antibiotic listed as sensitive. Daptomycin provides rapid bactericidal activity against this organism and is appropriate given the patient's normal renal function.

Antibiotic Drug History

Clinical Summary

Infection Summary

- **Type:** Recurrent lower-urinary-tract infection presenting as urethritis.
- **Likely pathogen:** *Streptococcus agalactiae* (Group B Streptococcus), a Gram-positive coccus.

Antibiotic susceptibility

- No predicted resistance.
- Predicted sensitive agent: **Daptomycin** (the only listed susceptible drug).

Recommended therapy

- **Daptomycin 6 mg/kg IV once daily** (e.g., 480 mg IV over 30 min for an 80-kg patient).

Rationale

- *S. agalactiae* is highly susceptible to daptomycin, which provides rapid, concentration-dependent bactericidal activity against Gram-positive organisms.
- The patient's renal function is normal (serum creatinine 0.7 mg/dL), allowing standard dosing without adjustment.

Precautions & monitoring

- Obtain baseline creatine phosphokinase (CPK) and repeat weekly; discontinue if CPK exceeds $5 \times \text{ULN}$.
- Watch for hypersensitivity reactions.
- Daptomycin is not indicated for pulmonary infections; ensure the infection remains confined to the urinary tract.

Considerations

- Prior use of azithromycin and clindamycin is not expected to affect daptomycin efficacy.
- Continue standard supportive care and advise on risk-reduction measures (e.g., proper genital hygiene, safe sexual practices).